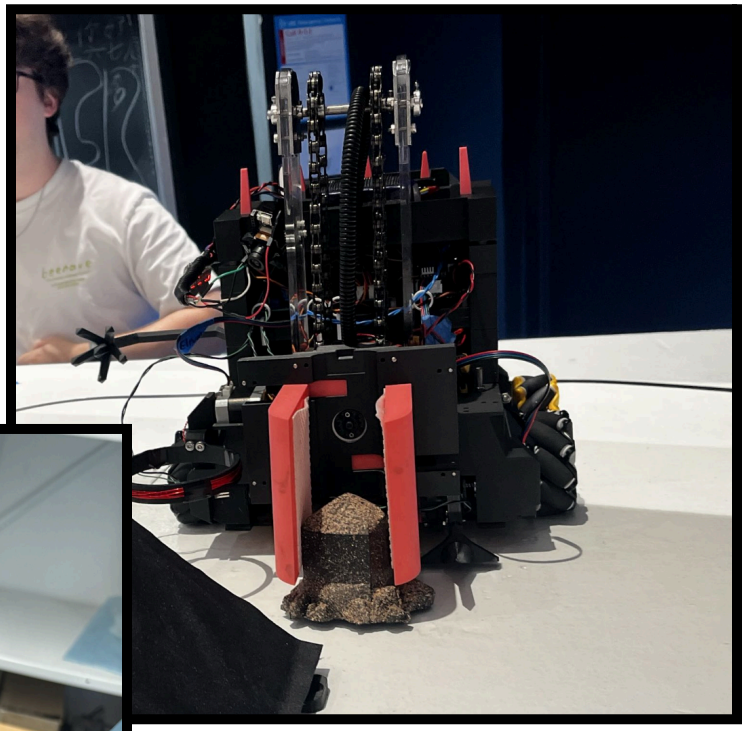
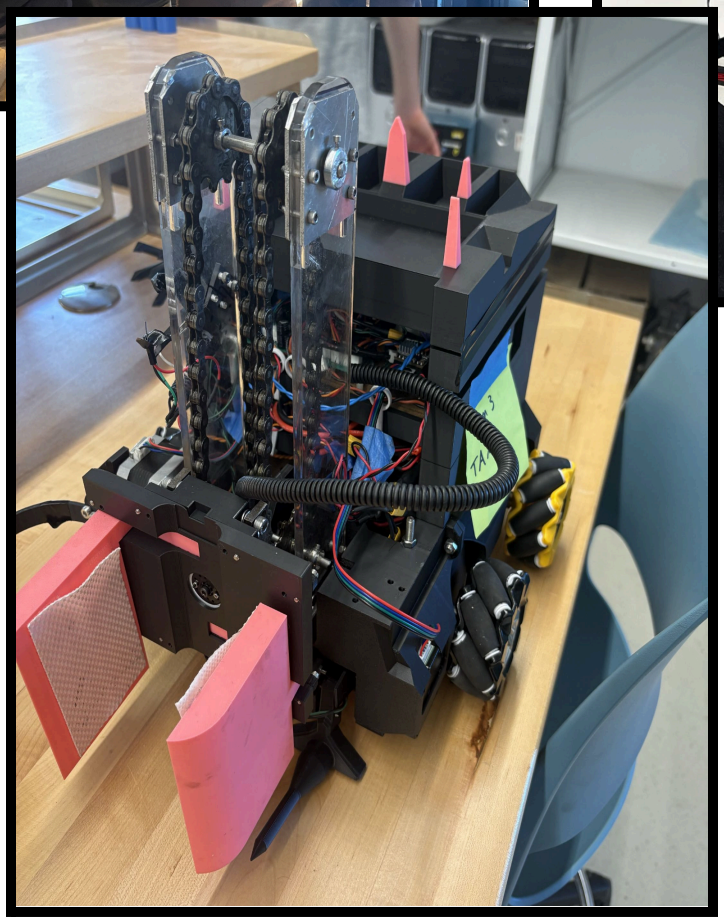
**Autonomous Robot Competition - May - July 2026**

Working in a team of four, I designed, fabricated, and programmed an autonomous robot from scratch for the UBC ENPH 253 design competition. Our robot secured 5th place overall by successfully integrating mechanical design, embedded electronics, and software control under strict competition constraints. I spearheaded the mechanical design, creating CAD models of many of the robot's components inside our team's OnShape. I also did a significant amount of machining, cutting out steel gears on the waterjet, frame pieces on a laser cutter, and shaft collars on a mill and lathe. I took on various additional work, including designing a custom PCB with KiCad for power distribution, and programming the routing for the robot. Our team was one of the only ones in the competition's history to use chains, which we needed for our unique competition strategy. While they proved to be quite a challenge to work with, I learned a lot about mechanical design from them, such as overfitting and tensioning systems.



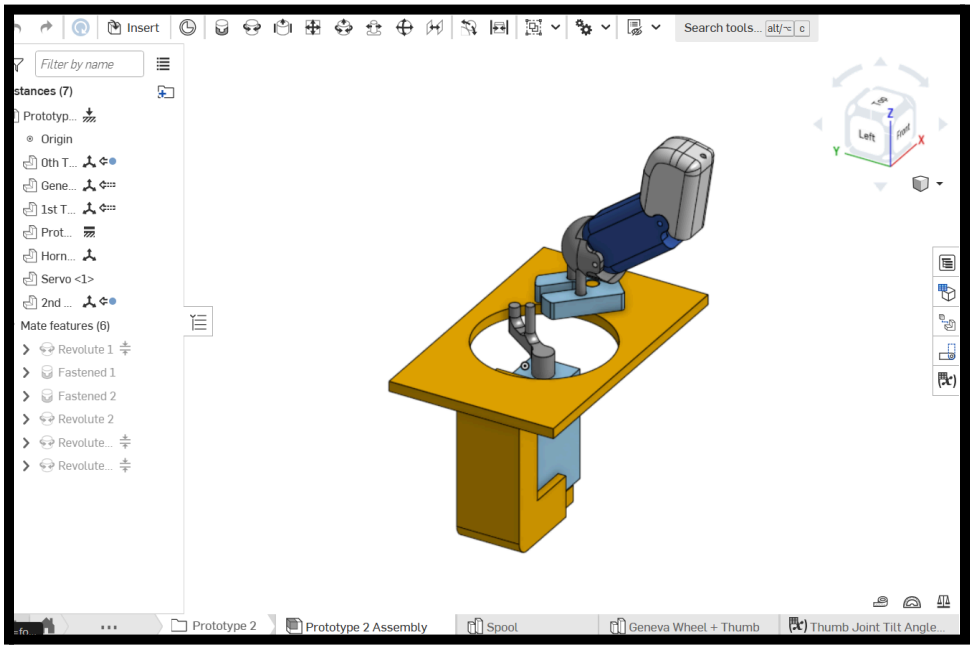




**UBC Bionics - September 2025 - Present**

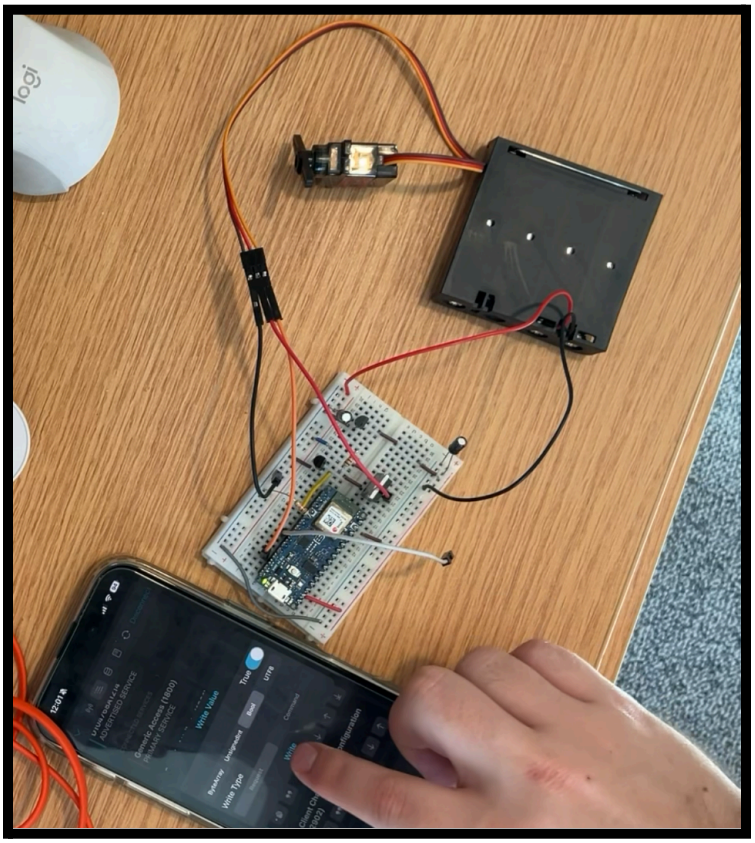
As part of the mechanical team on UBC Bionics, I have worked on developing a prosthetic arm to be used by amputees. Most of my work has centered on the thumb and fingers. As a senior member, in 2026 I took on the challenge of fully redesigning the thumb's actuation method, allowing for a two-stage system. Working with a limited amount of room for motors and electronics while maintaining a strict anthropomorphic design has forced me to be creative and expand my skillset as an engineer. I am constantly researching better techniques and discovering new mechanisms to try. In my first year with the team we worked exclusively in SolidWorks, but have recently switched to OnShape with strict version control rules.

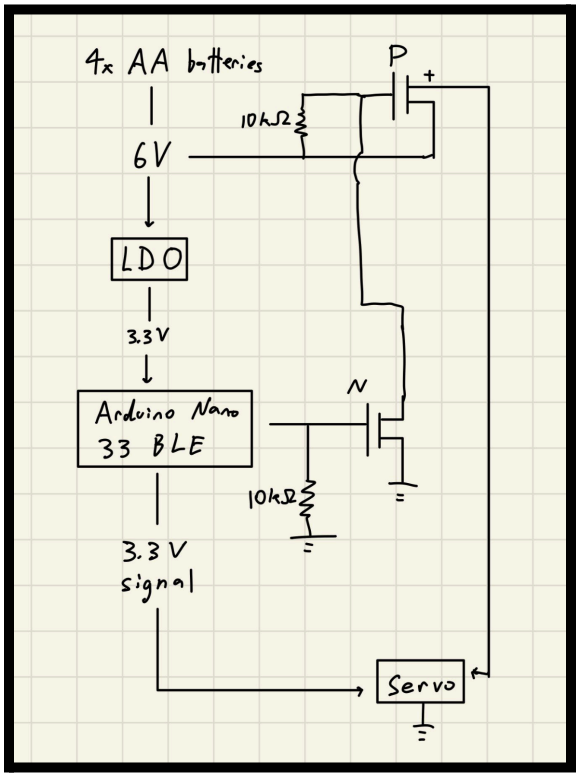




**Low Power Servo Controller - August 2026**

After becoming frustrated with my apartment's lack of remote-controlled light switches, I decided to attempt to design one of my own. I designed a schematic for the hardware, implementing a level-shifted high-side switch to prevent the servo from drawing power continuously. The goal was to optimize for power, so the batteries would rarely need to be replaced, so I looked for components with low quiescent power draw. To finish, I used a shunt-resistor to measure current flow and identify errors and sources of excess power draw. I then compared this to the current flow with no optimizations (standard components, no servo switch), and saw a 19x reduction! In the future, I would like to design the casing to attach it to my light switch panel.





**Website - August 2024**

I learned the basics of web design and backend data management in order to develop a site to run friendly bets with my friends on a professional frisbee league in the US. I used Firebase to host it, and learned Javascript, React, and CSS for the website itself.

